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Paleomagnetism and Age Correlation of the Mesoproterozoic Rocks of the Udzha and Olenek Uplifts, Northeastern Siberian Platform

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Abstract—This paper presents the results of paleomagnetic studies of the Proterozoic sedimentary and igneous rocks composing the structure of sections of the Olenek and Udzha uplifts. Carbonate rocks of the upper unit of Khaipakh formation have been studied within the Olenek uplift, and terrigenous and volcanic-sedimentary rocks of the Udzha and Unguokhtakh formations and the Early Mesoproterozoic igneous complex have been investigated within the Udzha uplift. Within the Olenek uplift, the carbonate rocks of the upper unit of the Khaipakh formation have been studied; within the Udzha uplift–terrigenous and volcanogenic-sedimentary rocks of the Udzha and Unguokhtakh formations, and the Early Mesoproterozoic magmatic complex. The results of the paleomagnetic study (1) indicate that the upper unit of Khaipakh formation of the Olenek uplift and the Udzha formation of the Udzha uplift have different sedimentation age, which contradicts the currently accepted correlation scheme of the Proterozoic sections in the northern Siberian platform; (2) support the existence of two stages of the Proterozoic magmatism in the territory of the Udzha uplift, with a more ancient stage having an age of ~1500 Ma according to the paleomagnetic data; (3) show that from ~1500 to 1110 Ma ago, the Siberian platform was located in the equatorial and subequatorial latitudes.

Keywords: paleomagnetism, Siberian platform, Riphean, Mesoproterozoic, Proterozoic, stratigraphy, paleomagnetic pole

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INTRODUCTION

A paleomagnetic method is a powerful tool in solving a broad variety of issues in the Earth sciences overall and in the Precambrian geology particularly. The results of the paleomagnetic studies play a significant role in paleotectonic and paleogeographic reconstructions, in studying the supercontinental cycles, in defining and correlating the geological events, in dating the age and formation duration of rocks, and in solving a plethora of other geological problems.

Among today's challenges faced by the study of paleomagnetism of the Siberian Platform is the plotting of a detailed trajectory (a curve) of the apparent polar wander path (APWP) for the platform. To date, this curve has been developed to a varying extent for the different time periods. There are segments for which there are still extremely few paleomagnetic poles known, which significantly limits the possibilities of plotting this curve. These intervals include, in particular, the time span between 1600 and 1000 Ma which corresponds to the Mesoproterozoic on the International geochronological timescale (Cohen et al., 2013). Despite the fact that in recent years a number of important paleomagnetic determinations have been obtained for this interval, which more or less meets the modern requirements for reliability and quality (Evans et al., 2016 and references therein), the number of available paleomagnetic poles is still insufficient for the confident tracing of the Mesoproterozoic segment of the Siberian APWP. The small amount of available poles is largely due to the small number and extreme inaccessibility of objects favorable for making the corresponding paleomagnetic determinations.

This study aims at obtaining new Mesoproterozoic paleomagnetic poles for the Siberian Platform based on studying the sedimentary and igneous rocks of the Udzha and Olenek uplifts. The choice of these objects for conducting the paleomagnetic studies was motivated by the existence of the reconnaissance data and the results of the previous research which show that these objects are promising and worth a detailed study on the up-to-date methodical level (Gurevich, 1983; Rodionov, 1984; Konstantinov et al., 2007; Pavlov et al., 2015).

GEOLOGICAL OUTLINE OF STUDY OBJECTS

The Udzha and Olenek uplifts predominantly composed of the Precambrian rocks are located in the

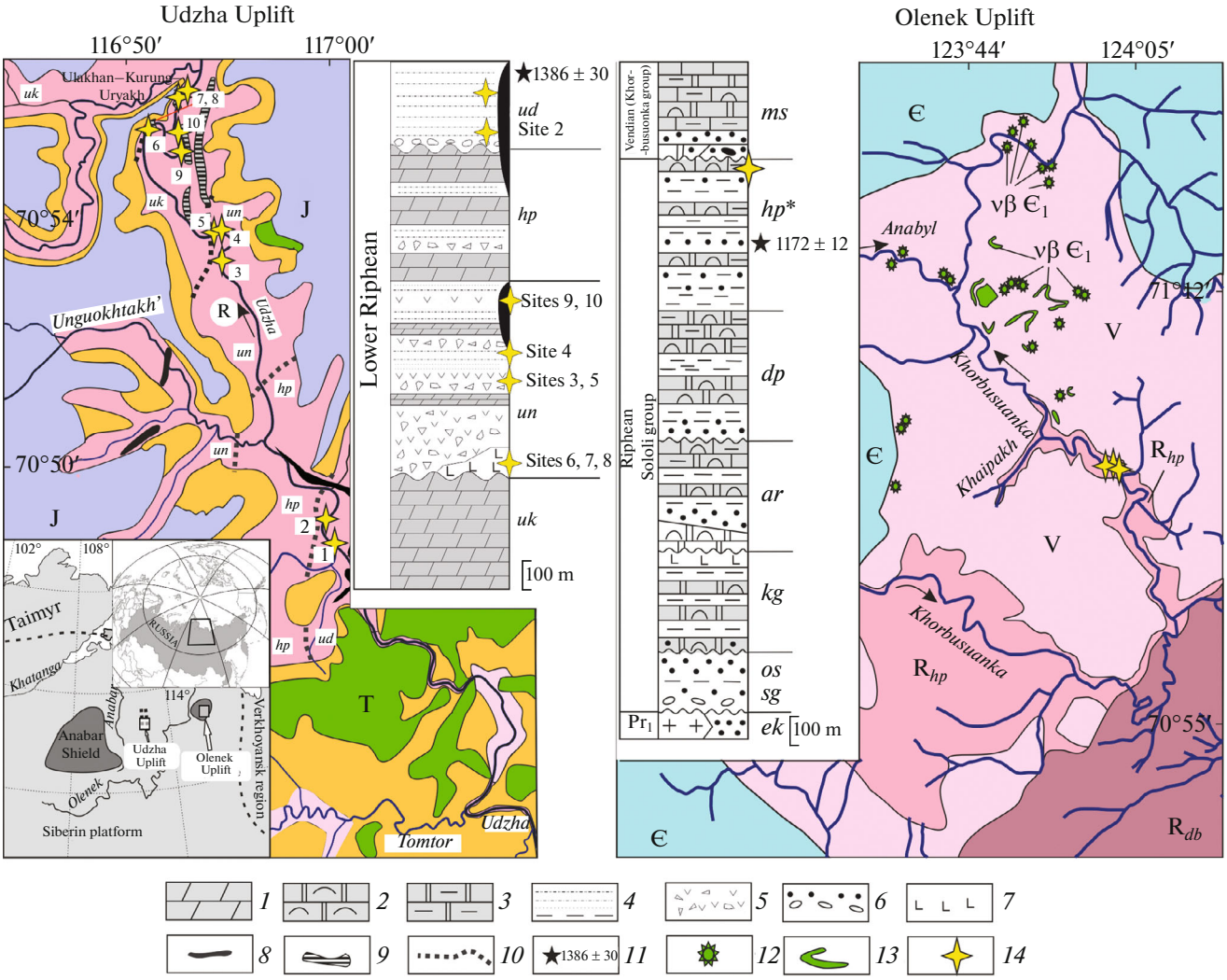


Fig. 1. Geological schemes of Udzha and Olenek uplifts according to (Gladkochub et al., 2009; *Ob"yasnitel'naya*..., 2016). Formations of Udzha Uplift: *uk*, Ulakhan–Kurung; *un*, Unguokhtakh; *hp*, Khapchanyr; *ud*, Udzha. Formations of Olenek Uplift: *ek*, Eekit; *sg,* Sygynyakhtakh; *os*, Osorkhayat; *kg*, Kyutyungda; *ar*, Arymas; *db*, Debengda; *hp**, Khaipakh; *ms*, Maastakh. Legend: (*1*) limestone, dolomite; (*2*) stromatolite limestones and dolomites; (*3*) sandy dolomites; (*4*) mudstones, siltstones, sandstones; (*5*) tuffs, tuff conglomerates, tuff breccias; (*6*) conglomerates, sandstones; (*8*) sheet bodies of basic composition; (*8*) intrusive bodies of stage II of Udzha Uplift magmatism; (*9*) intrusive bodies of stage I of Udzha Uplift; (*10*) boundaries of Proterozoic formations of Udzha uplift; (*11*) age of rock, Ma; (*12*) diatremes of Olenek Uplift; (*13*) sheet magmatic formations; (*14*) location of paleomagnetic sites in plan and in section.

northeast of the Siberian platform in the Udzha river valley (the right tributary of the Anabar River) and in the valleys of the Olenek river and its tributaries, respectively (Fig. 1).

Udzha Uplift

The Proterozoic part of the Udzha Uplift section is constituted of four formations (from bottom to top along the section): the terrigenous-carbonate Ulakhan-Kurung, the volcanic-sedimentary Unguokhtakh, the terrigenous-carbonate Khapchanyr, and the predominantly terrigenous Udzha (Shpunt et al., 1976). In this work we will present the results of the paleomagnetic study of the Unguokhtakh and Udzha formations.

According to (Okhlopkov et al., 1987), the Unguokhtakh Formation is subdivided into three units composed of tuffs, siltstones, and schists. Sampled for the paleomagnetic studies was the 100-m thick middle unit represented at the base by tuff breccias and tuff conglomerates which are changed, up the section, by red siltstones with substantial amount of carbonate concretions (Fig. 2).

The Udzha Formation includes two units and both were sampled for the paleomagnetic analysis. The lower unit with a thickness of 150 m is composed of variegated tephrogenic siltstones and mudstones with

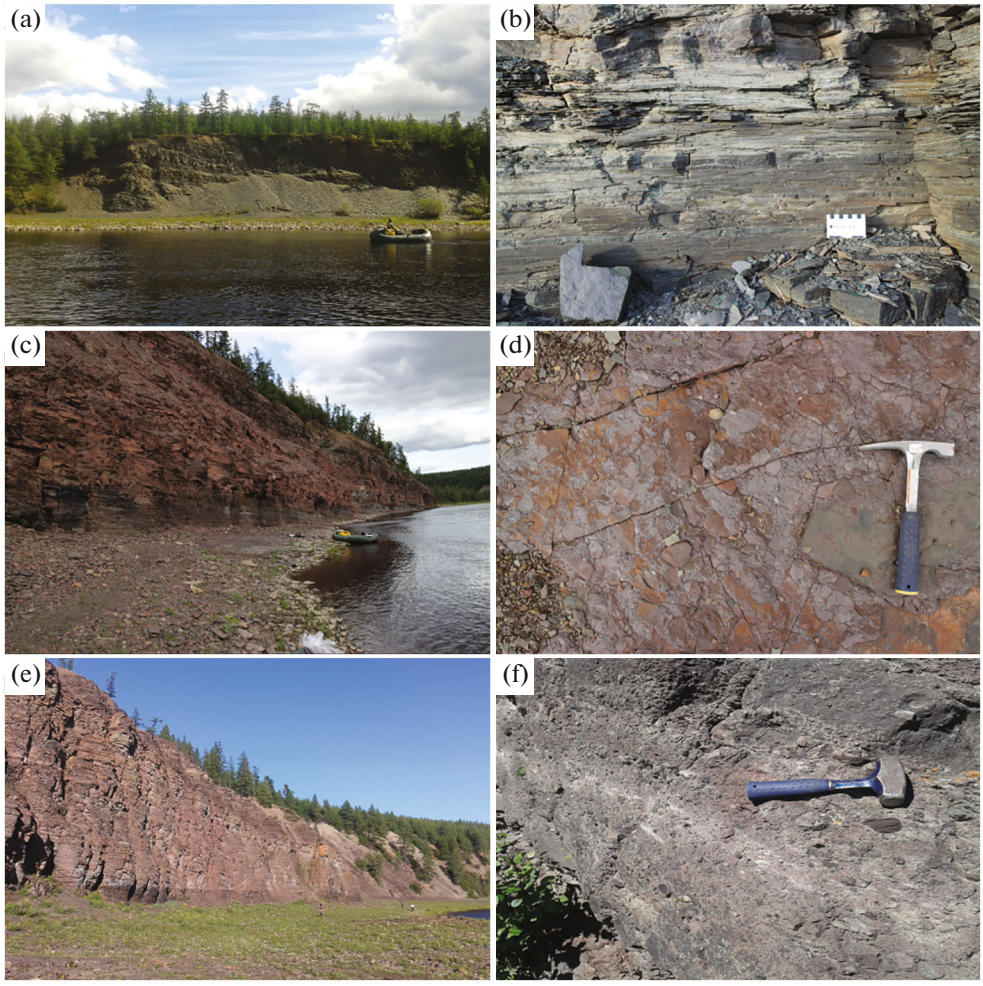


Fig. 2. Udzha and Unguokhtakh formation rock outcrops: (a) general view of outcrop of upper unit of Udzha formation sandstones; (b) sandstones of upper unit of Udzha formation; (c) general view of outcrop of lower unit of Udzha formation sandstones and siltstones; (d) pebble of intraformational conglomerates of lower unit of Udzha formation; (e) variegated siltstones of site 4 of Unguokhtakh Formation; (f) tuff conglomerates of site 3 of Unguokhtakh Formation.

interbeds of fine-pebble intraformational conglomerates (Fig. 2). The upper unit (50 m thick) is constituted of variegated tabular tephrogenic sandstones, gravelites and siltstones.

Within the Udzha uplift, the Proterozoic sedimentary rocks have mainly monoclonal bedding with layers dipping to the south and southeast at the angles of 5°–15°.

It has long been believed that the rocks of the Precambrian part of the Udzha section were formed within the Late Mesoproterozoic–Neoproterozoic (Semikhatov and Serebryakov, 1983). This opinion relied on the results of interregional correlation of stromatolite forms and the individual age determinations of K–Ar bulk dating of basite intrusions, cutting through the sedimentary rocks of the section, and basalts at the base of the Unguokhtakh Formation (from 1300 to 840 Ma (Osipova and Porshnev, 1966; Shpunt et al., 1982)). Based on these data, the Udzha Formation was correlated to the Neoproterozoic (Semikhatov and Serebryakov, 1983) and seemed to be a promising object for obtaining the Siberian Neoproterozoic paleomagnetic pole. However, the recent studies yielded new geochronological results indicating the Mesoproterozoic age of the Udzha Formation (Malyshev et al., 2018).

Based on the petrographic and petrochemical criteria, the magmatic bodies of the Udzha uplift are divided into two groups (Shpunt et al., 1976; Gladkochub et al., 2009). The first group of bodies cuts through the lower part of the section (the Ulakhan-Kurung and Unguokhtakh formations) and is interpreted as a product of magmatism at the initial stages of intracontinental rifting. The second group of bodies penetrates, besides the other formations, also the Khapchanyr and Udzha formations and is related to more recent stages of the evolution of rift systems (Shpunt et al., 1979; Gladkochub et al., 2009), which

may indicate its substantially younger age compared to the first group. For the second group, the following datings were obtained: 1074 ± 11 Ma (Ar–Ar, plagioclase) and 1386 ± 30 Ma (U–Pb, apatite) (Gladkochub et al., 2009; Malyshev et al., 2018).

In our paleomagnetic studies, we sampled the magmatic bodies pertaining to both stages of the Precambrian magmatism of the Udzha uplift. The results of paleomagnetic study for the rocks of the second stage (~1385 Ma) are presented in (Malyshev et al., 2018). The igneous rocks of the first stage are studied in this work. In our study, we assume an older age for the second group of the bodies (~1385 Ma) due to the fact that this age determination was obtained from a more "stable" isotope system. The coincidence of the paleomagnetic directions obtained for the bodies of the second stage of magmatism and the directions obtained for the Chieres dike of the Anabar massif (U–Pb, baddeleyite, 1384 ± 2 Ma) (Ernst et al., 2000; Malyshev et al., 2018) is yet another argument supporting this age option.

Olenek Uplift

Within the Olenek uplift, we studied the Khaipakh Formation. This formation overlays the Mesoproterozoic dolomites of the Debengda Formation and is composed of the alternating carbonate (dolomites, stromatolite dolomites, and marls) and terrigenous rocks (sandstones, mudstones) with interbeds of greenish and reddish varieties. In the survey region, the studied rocks of the Khaipakh Formation have subhorizontal bedding. Two sedimentation hiatuses detected within the Khaipakh Formation divide it into three units (Shpunt et al., 1982; *Ob"yasnitel'naya*..., 2012). Various stromatolite associations are noted in each unit. The change in the stromatolite associations of the lower and middle units guides the boundary between the Meso- and Neoproterozoic ~1030 Ma (Semikhatov et al., 2000; Zaitseva et al., 2017 and references therein). Thus, the middle and upper parts of the Khaipakh Formation have been until recently attributed to the Neoproterozoic while the lower part has been referred to the Mesoproterozoic. The Mesoproterozoic age of the lower part of the Khaipakh Formation is suggested by the K–Ar and Rb–Sr dating of glauconites sampled from this stratigraphic interval (1172 ± 18 and 1112 ± 24 Ma) (Zaitseva et al., 2017). Unfortunately, the younger age limit of the Upper Khaipakh Formation can only be constrained by the datings of detrital zircons from the overlying Maastakh Formation (~630 Ma) (Vishnevskaya et al., 2017).

According to the schemes of interregional correlation based on similar stromatolite forms (Semikhatov and Serebryakov, 1983; *Resheniya*..., 1983), the upper unit of the Khaipakh formation is correlated in terms of the age with the Udzha Formation of the Udzha Uplift. However, if this correlation is correct, then, based on the age constraints for the Udzha Formation (>1385 Ma) obtained in our previous studies (Malyshev et al., 2018), the upper unit of the Khaipakh formation cannot be Neoproterozoic. But if it indeed has a Neoproterozoic age, then the correlation of this unit to the Udzha Formation should be admitted wrong. The results of paleomagnetic studies of the Udzha Formation and upper unit of the Khaipakh formation may provide additional arguments in favor of one of these assumptions.

ROCK SAMPLING AND LABORATORY PROCEDURES

Rock Sampling

For solving the tasks of our study, we carried out paleomagnetic sampling of both the sedimentary and magmatic bodies of the Udzha and Olenek uplifts. The studied objects and the sampling coordinates are presented in Table 1.

The sedimentary rocks of the Udzha Formation (sites 1 and 2) were sampled from the bottom upwards along the section. Oriented samples were taken with an average interval of 0.5–3.0 m predominantly from red and fine-grained lithological varieties. Overall, a total of ~150 m of the section was sampled. For paleomagnetic reliability test, pebble samples were taken from the 2–10 cm thick interbeds of intraformational conglomerates.

Samples of the Unguokhtakh siltstones (site 4) were taken from bottom upwards along the section with an interval of 0.5–1.0 m. Tuff and tuff conglomerate samples (sites 3 and 5) were taken at random from each particular outcrop; both the tuffaceous matrix and rock chips, mainly basalts and cherry siltstones, were sampled. Overall, sampling covered a total of ~100 m of the section of the Unguokhtakh Formation.

The sills (sites 9 and 10) were sampled uniformly along the strike of the body with sampling of both the endocontact and central parts of the sill. Five samples of the hosting gray carbonates were taken at a distance of 5–60 cm from the contact. Sampling the host rocks at a substantial distance from the contact was impossible because these rocks are not exposed.

The rest of the magmatic bodies (sites 6, 7, 8) were sampled by the site-based sampling method. Before orienting the samples of magmatic bodies by a geological compass, the latter was checked for the absence of influence of tested rocks on its needle.

The Khaipakh Formation in our collection is represented by the samples of the upper unit of the Khaipakh formation. These samples were taken from the relatively narrow bedrock outcrops of stromatolite limestones exposed in the Horbusuonka river valley at a distance of 100–500 m from each other; each outcrop was considered a separate site (sites 2a, 2b, 2c, S14OLD, and S15OLD). The sites were sampled from bottom upwards along the section with an interval of

Table 1. Objects of paleomagnetic studies and coordinates of sampling sites

Object		Site no.	Rock	Number of samples	Site coordinates
Khaipakh Formation	Upper unit	2a	Stromatolite limestone	41	N 71°05.281′ E 124°02.294′
	Upper unit	2b		7	N 71°05.273′ E 124°02.165′
	Upper unit	2c		10	N 71°05.267′ E 124°02.050′
	Upper unit	S14OLD		10	N 71°05.135′ E 124°02.922′
	Upper unit	S15OLD		10	N 71°05.135′ E 124°02.922′
Udzha Formation	Upper unit	1	Siltstone, sandstone	35	N 70°48.863′ E 117°00.544′
	Lower unit	2		57	N 70°49.325′ E 116°56.864′
Unguokhtakh Formation	Middle	3	Tuff conglomerate	23	N 70°53.619′ E 116°54.782′
	Middle	4	Siltstone	57	N 70°53.810′ E 116°54.933′
	Middle	5	Tuff conglomerate	7	N 70°53.828′ E 116°54.084′
Magmatic bodies	Flow	6	Basalt	14	N 70°55.476′ E 116°51.993′
	Flow	7	Basalt	7	N 70°55.652′ E 116°52.360′
	Dike	8	Carbonatite	9	N 70°55.879′ E 116°52.502′
	Sill	9	Dolerite	33	N 70°55.115′ E 116°52.686′
		10		24	N 70°55.346′ E 116°52.796′

0.5–1.5 m, with seven to 41 samples taken from each site.

We have sampled three sites of the upper unit of the Khaipakh formation. The processing data for the samples taken in 2001 from yet another two sites by V.E. Pavlov and A.V. Shatsillo, the employees of the paleomagnetic laboratory of Schmidt Institute of Physics of the Earth of the Russian Academy of Sciences (IPE RAS), were also analyzed in this work.

Laboratory Procedures

The collections were processed in the Laboratory of the Main geomagnetic field and rock magnetism of IPE RAS. The processing included preparation of oriented samples for measurements, a series of rock magnetic and paleomagnetic experiments.

The complex of the rock magnetic studies was conducted on a thermomagnetometer designed by Yu.K. Vinogradov (Geophysical Observatory (GO) "Borok"), three-component vibration magnetometer (GO "Borok"), PMC MicroMag 3900 vibration magnetometer (Lake Shore, USA) and MFK1-FA kappa-bridge with CS3 temperature control unit (AGICO, Czech Republic).

The rock magnetic parameters were interpreted with the use of the Day–Dunlop diagram (Day et al., 1977; Dunlop, 2002a; 2002b). The coercivity spectra of samples were analyzed by the method of cumulative log-Gaussian analysis (CLGA) of normal magnetization curve (Kruiver et al., 2001) in the MAX UnMIX online application (Maxbauer et al., 2016).

Microprobe studies were carried out at GO "Borok" on Tescan Vega II scanning electron microscope.

The remanent magnetization measurements during thermal demagnetization were carried out on a cryogenic (SQUID) magnetometer (2G Enterprises, USA) with automatic sample changer and on JR-6 spin magnetometer (AGICO, Czech Republic) in a space shielded from the external magnetic field by a system of Helmholtz coils or in a non-magnetic room with the 200-fold reduction of the ambient magnetic field in the interior. Stepwise temperature demagnetization of the samples was conducted in MMTD80 nonmagnetic furnace (Magnetic Measuremenets, Great Britain) with the uncompensated field of at most 5–10 nT. The overall number of demagnetization steps varied from 10 to 20.

The demagnetization results were processed by the standard procedure (Kirschvink, 1980; Khramov et al., 1982) using Enkin's computer program package (Enkin, 1994). For graphical representation of the material for paleoreconstructions we used the GMAP2003 program (Torsvik and Smethurst, 1999).

The parameters of the geomagnetic field in the study region were determined from the IGRF-12 model (Thébault et al., 2015).

RESULTS OF PALEOMAGNETIC AND ROCK MAGNETIC STUDIES

Udzha Formation

The value of natural remanent magnetization (NRM) in the studied rocks of the Udzha Formation varies within 0.001–0.3 A/m. The component analysis revealed four magnetization components.

The low-temperature magnetization component (LT) is destroyed at temperatures up to 180–250°C and is probably recent since its direction coincides with the direction of the present magnetic field in the study region (Fig. 3).

In 35 samples, a bipolar high-temperature component (*HT*) with southwestern or northeastern declination and predominantly low positive inclination is isolated in the temperature interval from 400–500 to 680°C. This component is probably associated with hematite and/or magnetite (Fig. 3).

The presence of hematite (or Ti-poor hemoilmenite) is supported by the results of thermomagnetic analysis which have a sharp drop in saturation magnetization *Is* in the temperature interval 650–680°C (Fig. 3) and by the values of remanent coercive force determined in the samples of the formation which are above 400 mT (Fig. 5a). At the same time, the presence of the Hopkinson peak at 550–580°C in the temperature dependence of magnetic susceptibility $\kappa(T)$ (Fig. 3) suggests that the samples of the Udzha Formation can also contain magnetite or Ti-poor titanomagnetite. This conclusion is supported by the CLG-analysis (cumulative log-Gaussian, (Kruiver et al., 2001; Maxbaueret al., 2016)) of the normal magnetization curve which, in addition to the highly coercive mineral, also established the presence of a low-coercive mineral with a median saturation field $H_{1/2}$ = 63 mT (Figs. 5b and 5c) in the sandstone sample of the Udzha Formation.

Twenty samples of the Udzha Formation contain, in addition to the *LT* and *HT* components, a medium-temperature magnetization component with unblocking temperatures in the interval 380–540°C. Among vector directions corresponding to this component there is a cluster with the northwestern declinations and moderate inclinations and a cluster with the eastern declinations and moderate-high inclinations (Fig. 3, Table 1).

Hereinafter, the direction and component corresponding to the northwestern cluster are referred to as *MTW*, and the direction and component corresponding to the eastern cluster, as *MTE*.

The geological situation does not allow of conducting the fold test; however, the higher stability of the *HT* component to heatings and its bipolarity make this component much more eligible for the role of primary magnetization component than the *MHW*/*MTE* components. Other arguments in favor of the primary origin of this magnetization component will be considered below.

Thermal demagnetization of pebbles of intraformational conglomerates reveals the same *HT* and *MTE* components that we observe in the undisturbed layers of the Udzha Formation (Fig. 3). Thus, the conglomerate test is negative result ($r = 0.76$ with a critical value of 0.50) (Khramov et al., 1982). Following (Butler, 1998), we note, however, that while a positive conglomerate test definitely suggests primary origin of the magnetization, a negative test only indicates that the magnetization was acquired after the formation of the intraformational conglomerate layer. Moreover, the magnetization can be created almost immediately after the sediment lithification at the stage of diagenesis (Butler, 1998). In this case, for solving our tasks, this magnetization on the geological time scale can be considered as primary (Kodama, 2012).

The conglomerate test indicates that the *HT* component detected in the Udzha Formation is mainly associated with authigenic hematite.

Unguokhtakh Formation Site 4 (siltstones)

The NRM values in the siltstone samples from outcrop 4 range from 0.002 to 0.4 A/m. Thermal demagnetization isolates several magnetization components in these rocks. Part of samples contains a low-temperature component (*LT*) which is removed at 250–300°C (Fig. 6) and is close to the direction of the present magnetic field (Fig. 7). This suggests a recent age of this component.

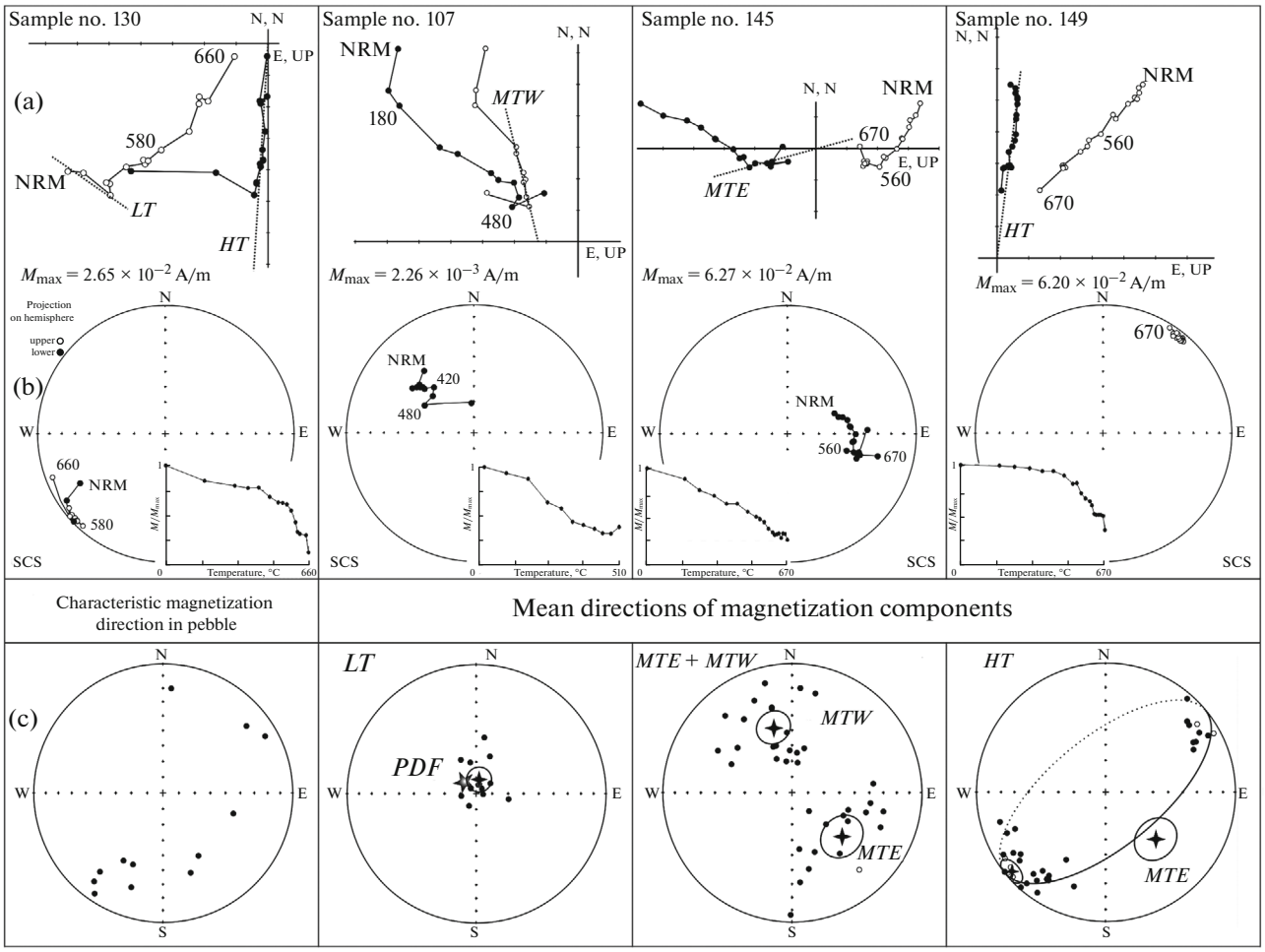


Fig. 3. Demagnetization results of Udzha Formation samples: (a), (b) representative Zijderveld diagrams and stereograms, respectively; (c) directions of isolated magnetization components in conglomerate pebbles. *PDF* is direction of present magnetic field in study region. Other acronyms and abbreviations are expanded in text. All diagrams are presented for stratigraphic coordinate system (SCS).

The medium-temperature component is isolated in the temperature interval from 300 to 540°C. The corresponding vectors are revealed in 14 samples with a rather noisy paleomagnetic record. Vectors have northwestern declinations and moderate positive inclinations and form a cluster on the stereogram (concentration parameter $K = 21.4$, Table 1) which is located approximately in the same region of the stereogram as the mean direction of the *MTW* component identified in the rocks of the Udzha Formation (Fig. 6).

It was established that 23 samples contain a stable high-temperature *HT* component with the northeastern declination and a moderate positive inclination (Fig. 6). The maximum unblocking temperatures of this component broadly vary among the samples from 480 to 660°C. This can be associated with the presence of different proportions of titanomagnetite, magnetite, and hematite in siltstone samples.

The presence of these magnetic minerals also follows from the results of rock magnetic studies. For example, the temperature curves $\kappa(T)$ for heating have Hopkinson peaks in the vicinity of 560–580°C whereas some samples have an additional peculiarity in the form of the bend or Hopkinson peak at temperature of about 500°C (Fig. 4).

The *Hcr* values on the order of 110–120 mT are characteristic of a mixture of magnetically relatively soft (magnetite, titanomagnetite) and hard minerals (hematite). This conclusion is supported by the behavior of the curve of normal magnetization which initially rapidly grows in the fields up to 100 mT then its growth increasingly slows down while not however ceasing even when the field reaches as high a value as 1.3 T (Figs. 5a, 5b). The decomposition of the coercive spectrum into components (Kruiver et al., 2001) indicates the presence of low- and high-coercive minerals in the sample (Fig. 5c). The median saturation fields

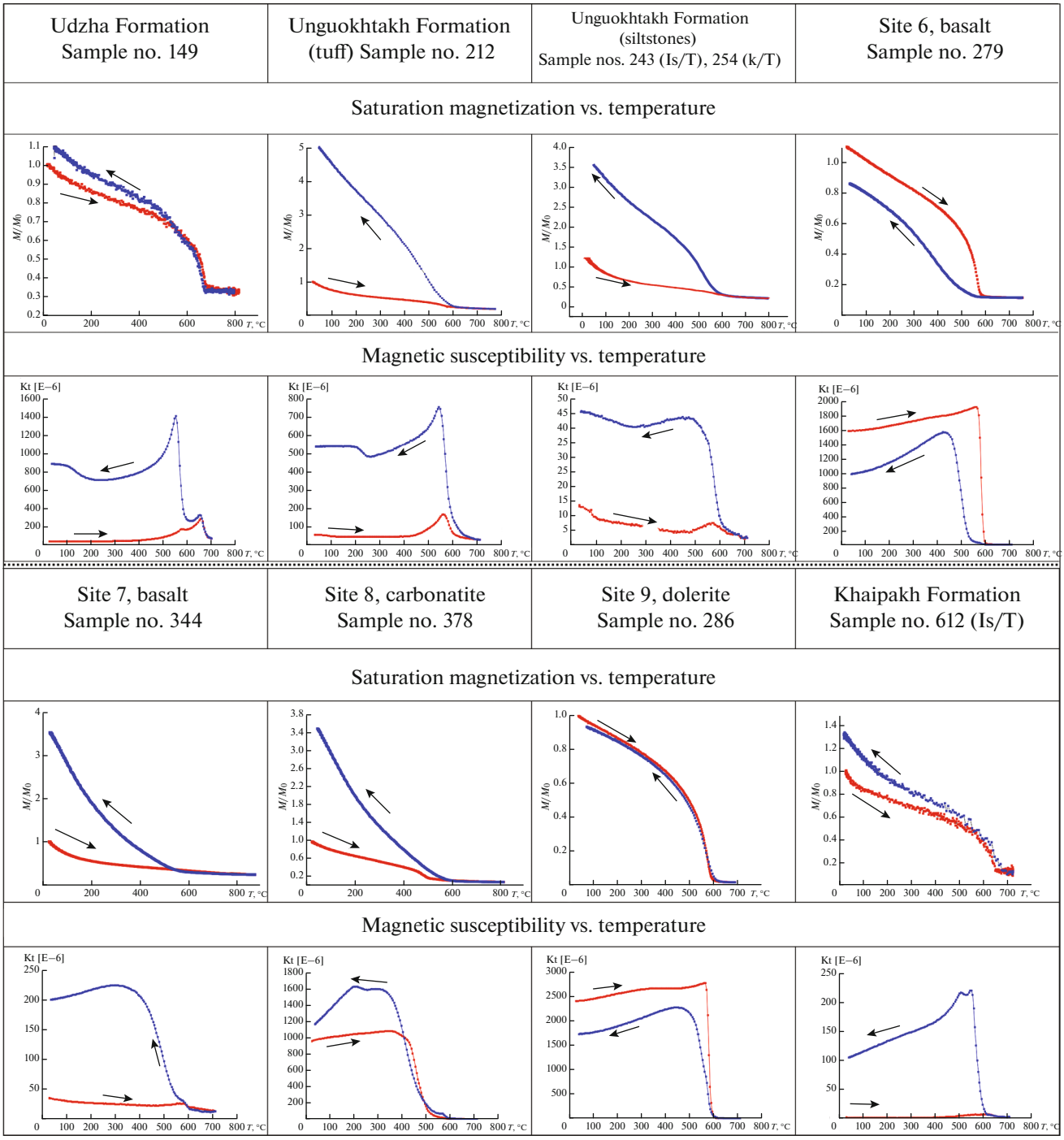


Fig. 4. Diagrams of dependence of saturation magnetization and magnetic susceptibility on temperature of magmatic and sedimentary rocks of Udzha and Olenek uplifts.

of these minerals are 38 and 260 mT which corresponds to magnetite and hematite, respectively.

Sites 3 and 4 (tuffs and tuff conglomerates)

The samples of these rocks as a rule contain a fairly distinct paleomagnetic signal, and *NRM* of the rocks varies within 0.008–0.1 A/m.

Besides the recent (as suggested by its direction) low-temperature *LT* component which is removed at the first demagnetization steps, also a high-temperature *HT* component with a southwesterly declination and a moderately negative inclination (Fig. 6) is observed. The maximum unblocking temperatures of this component which are located in the interval 580–600°C suggest indicate that this component is most

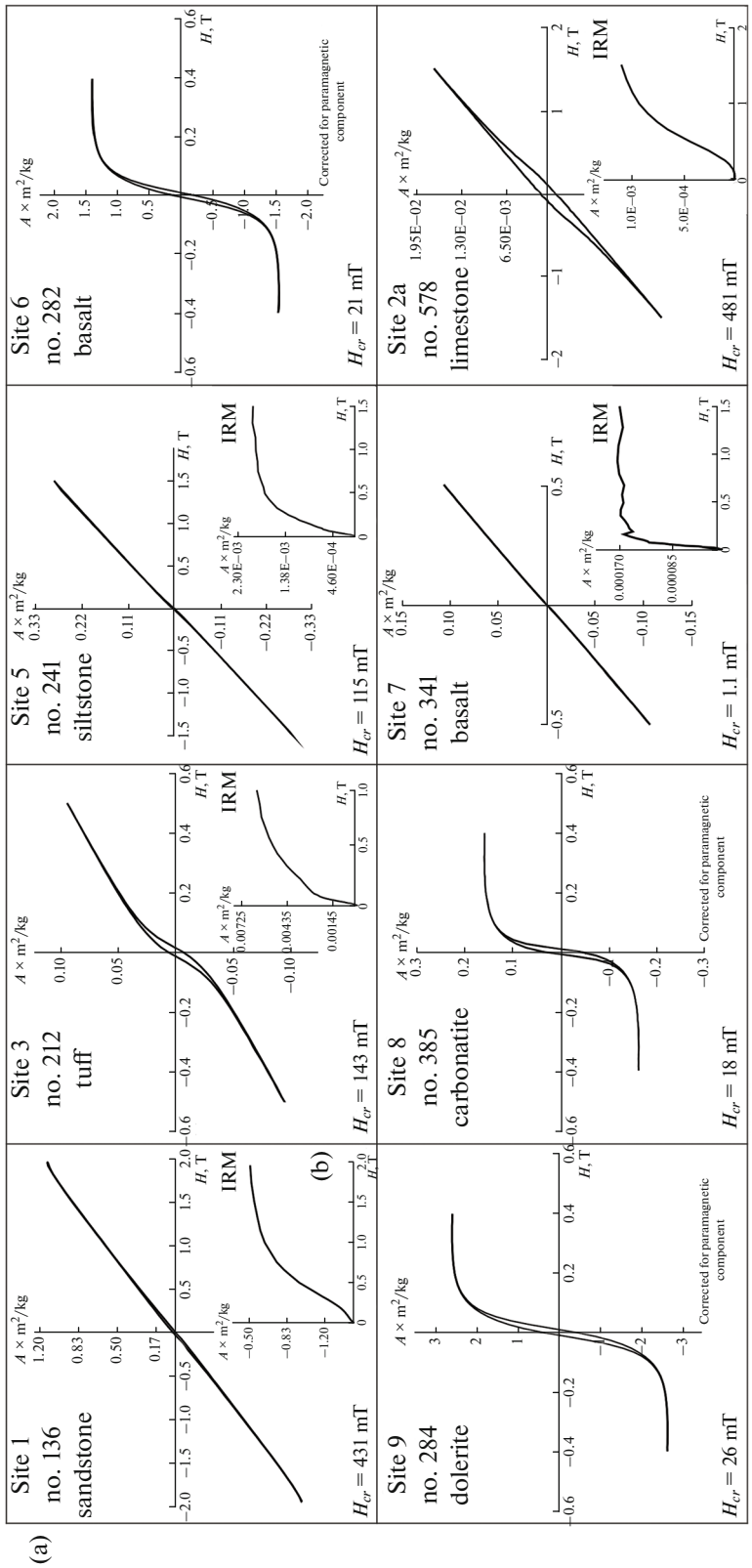


Fig. 5. Results of rock magnetic studies of sedimentary and igneous rocks of Udzha and Olenek uplifts: (a) hysteresis loops; (b) diagrams of normal magnetization of samples; (c) coercive spectra of magnetic minerals in samples. $H_{1/2}$ is median saturation field of magnetic minerals.

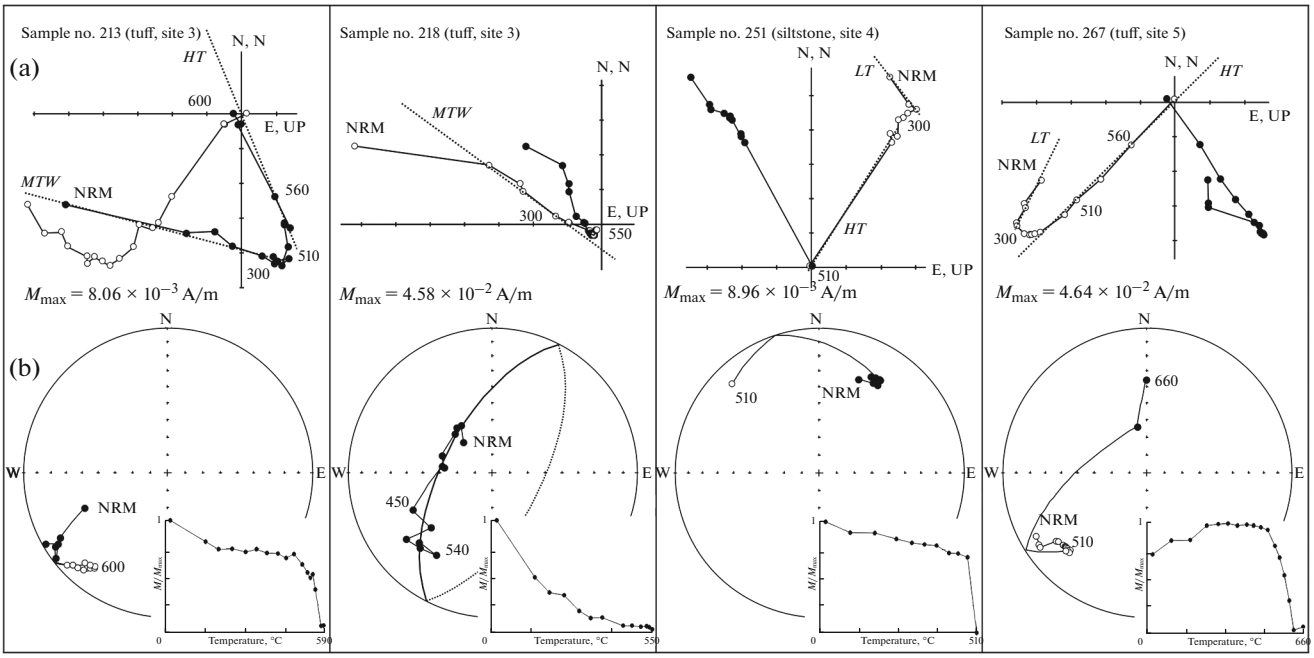


Fig. 6. Demagnetization results of Unguokhtakh Formation rocks: (a) representative Zijderveld diagrams; (b) stereograms and magnetization decay diagrams. Numbers in Zijderveld diagrams and stereograms show temperature in °C. Acronyms and abbreviations are expanded in text. Diagrams are presented for stratigraphic coordinate system.

likely carried by magnetite. In some samples, separating the *LT* and *HT* components is impossible because of the overlapping of the spectra of their unblocking temperatures. In this case, however, the remagnetization circles are confidently identified (Fig. 6, sample 218; Fig. 7, samples 217, 218, 233).

The conclusion that the main magnetic mineral in the tuffs of the Unguokhtakh Formation is magnetite (or Ti-poor titanomagnetite) is supported by the results of the rock magnetic studies. For example, the temperature dependence of saturation magnetization in these samples testifies to the presence of a magnetic mineral having the Curie points at 580–600°C (Fig. 4). The $\kappa(T)$ curve in the interval 560–580°C has the Hopkinson peak (Fig. 4), which is characteristic of the rocks containing single-domain or pseudo-single-domain magnetite (Butler, 1998). The presence of titanomagnetite grains in the tuffs of the Unguokhtakh Formation is confirmed by the results of microprobe analysis (see below).

Magmatic Bodies of the First Stage of the Udzha Uplift Magmatism Site 6 (basalts)

The NRM values in the basalt samples from site 6 vary within 0.02–0.7 A/m. The samples overall have a noisy paleomagnetic signal; however, individual samples contain a very high quality paleomagnetic record (Fig. 8a). The low-temperature component is not clearly pronounced. In most samples, the experiments fairly confidently established the presence of high-temperature component with the maximum unblocking temperatures from 510 to 580°C, with southwesterly declinations and moderate negative inclinations (Fig. 8a). The results of the thermomagnetic studies, the analysis of temperature dependence of magnetic susceptibility, and the hysteresis parameters indicate the presence of a magnetic mineral with the Curie temperature at about 580°C (Fig. 4) and saturation fields of 100–150 mT (Fig. 5), which is typical of magnetite.

Site 7 (basalts)

The NRM values in basalts from site 7 vary within 0.008–0.02 A/m. Based on the results of stepwise thermal demagnetization, the high-temperature magnetization component is isolated (Fig. 8b). The mean direction of the identified component has a southwesterly declination and moderate negative inclination and, in general, coincides with the direction of the high-temperature component in basalts described above. We note that the $Is(T)$ curves and the magnetic hysteresis loops for the samples from site 7 have a paramagnetic shape suggesting a fairly low content of ferromagnetic minerals and precluding from their identification (Fig. 5a). At the same time, the $\kappa(T)$ curve for the same samples has a peak and a bend in the vicinity of ~580°C (Fig. 4) testifying to the presence of magnetite.

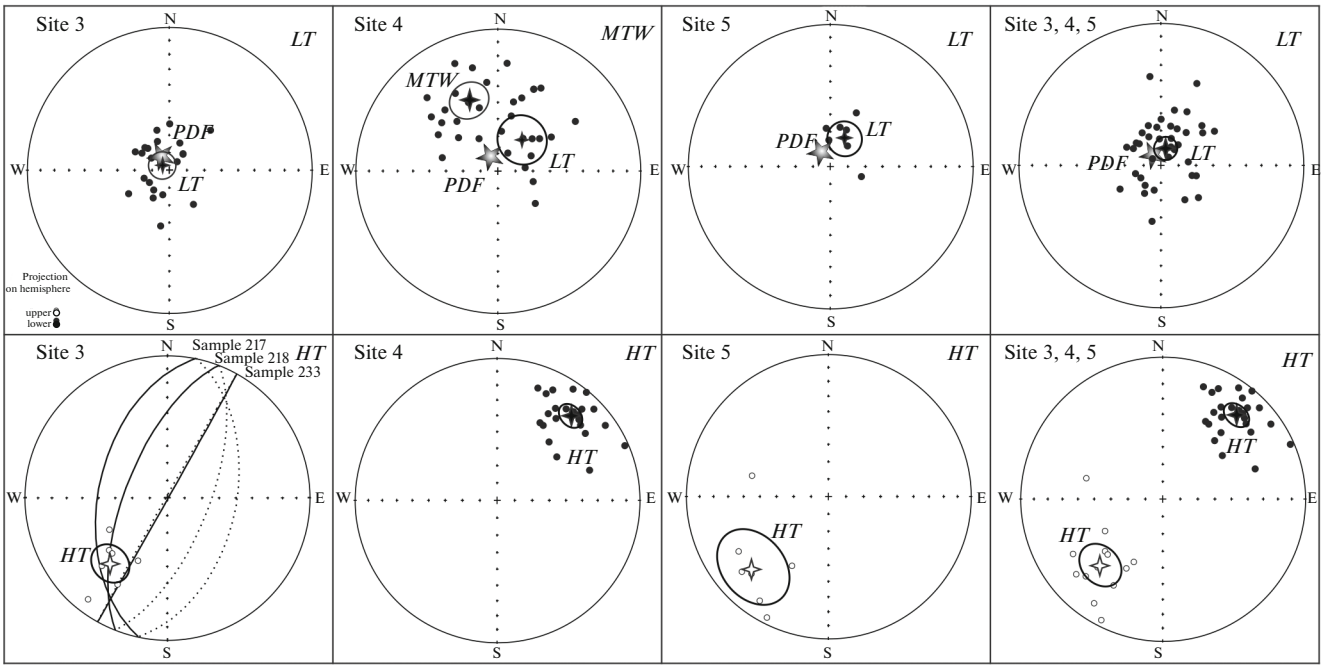


Fig. 7. Stereograms of directions of magnetization components isolated in Unguokhtakh Formation samples. *PDF* is direction of present magnetic field in study region. Other acronyms and abbreviations are expanded in text. Diagrams are presented for stratigraphic coordinate system.

Site 8 (carbonatites)

The NRM values in the samples of carbonatite dike vary within 1–2 A/m. The paleomagnetic record of the studied samples is strongly dominated by the high-temperature characteristic component with a northeasterly declination and low positive inclination and with the maximum blocking temperatures of 540–550°C (Fig. 8c). These temperatures are characteristic of the rocks where the main magnetic mineral is Ti-poor titanomagnetite. Curves $Is(T)$ and $\kappa(T)$ measured for the samples from the carbonatite dike also indicate the presence of a ferromagnetic mineral with the Curie points at about 520–540°C (Fig. 4), probably titanomagnetite with relatively low Ti content.

Sites 9 and 10 (dolerites)

The samples from both sites have NRM values within 0.5–2.0 A/m and a fairly noisy paleomagnetic signal similar to that described above.

The component analysis reveals two magnetization components: a low-temperature poorly stable component with the unblocking temperatures from 150 to 380°C and a stable high-temperature component (Fig. 9a). The vectors corresponding to the *LT* component have irregular directions. Some of them lie in the region the directions of the present geomagnetic field, and some are distributed chaotically (Fig. 9b).

The high-temperature magnetization component *HT* is isolated at temperatures 450–580°C. The values of the blocking temperatures, the analysis of the $Is(T)$ and $\kappa(T)$ curves whioch have a single bend in the interval of 580–590°C (Fig. 4), and the analysis of the hysteresis loops which reach saturation in the fields of 100–150 mT indicate the presence of magnetite (or Ti-poor titanomagnetite) in the studied samples (Fig. 5a).

The vectors corresponding to the *HT* component form two antipodal clusters of the directions with a southwesterly/northeasterly declination and a moderate negative/positive inclination on the stereogram (Fig. 9).

In the calculations of the mean direction of the vector of high-temperature magnetization component (Fig. 9c), the samples with a noisy paleomagnetic record ($\alpha 95 \geq 10°$) and the samples falling out of the distribution (angular standard deviation $\theta_{95} = 140/\sqrt{k}$ (Butler, 1998), were disregarded (Fig. 9d)).

It is worth taking a special note of the fact that in the Day–Dunlop diagram, most of the points corresponding to the considered igneous rocks of the Udzha Uplift lie in the field of the pseudo-single-domain grains in the region of the curve that marks the mixing of the single- and multi-domain particles (Fig. 10). This may indicate that in this case we are dealing with an ensemble of single-domain and multi-domain titanomagnetite/magnetite grains rather than with pseudo-single-domain particles alone.

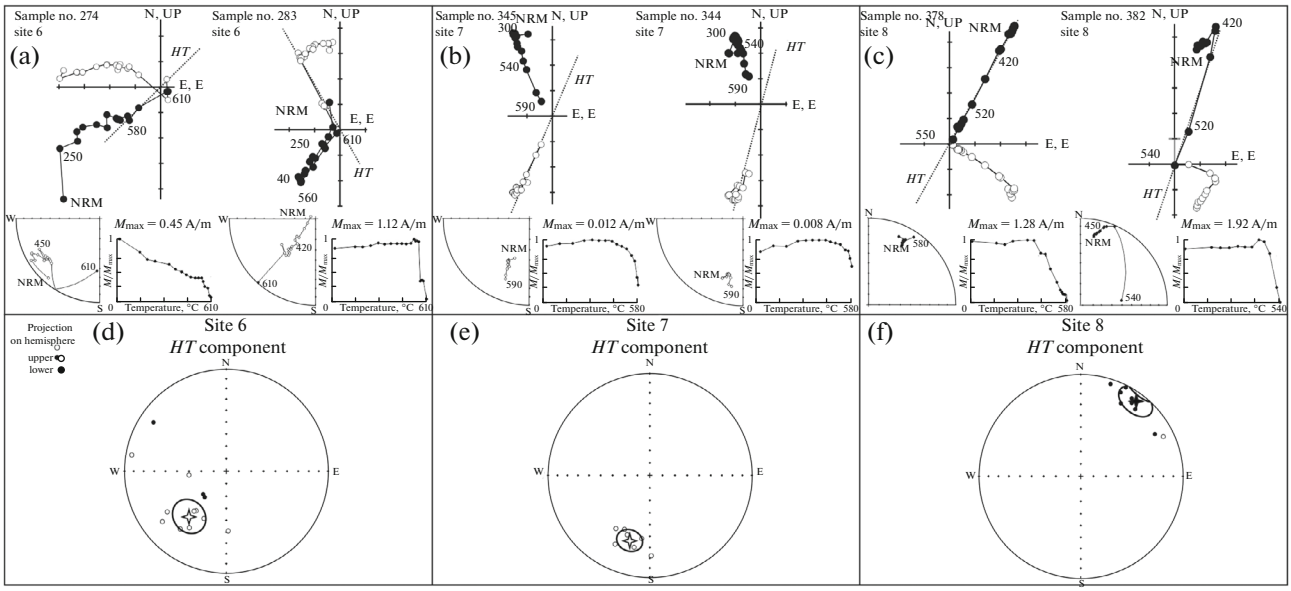


Fig. 8. Demagnetization results of Udzha Uplift magmatic bodies from sites 6, 7 and 8; (a), (b), (c) representative Zijderveld diagrams and stereograms; (d), (e), (f) directions of high-temperature magnetization components for each site. Acronyms and abbreviations are expanded in text. Diagrams are presented for stratigraphic coordinate system.

Olenek Uplift Khaipakh Formation (Upper Unit of the Khaipakh formation)

The studies have shown that the rocks carry a paleomagnetic signal of a varying quality. In the samples with a high-quality paleomagnetic signal, the magnitude of NRM vector varies within 0.001–0.02 A/m. In the rest of the samples making up ~ 40% of the collection, NRM is substantially lower dropping to 0.0001 A/m.

Thermal demagnetization of the studied samples isolated two magnetization components (Fig. 11): a medium-temperature component (*MT*) with the unblocking temperatures of 250–520°C and a high-temperature component (*HT*) with the unblocking temperatures 560–700°C.

A probable explanation of such an extended interval of unblocking temperatures is that the magnetization is carried by hematite grains with broadly varying grain sizes (Jiang et al., 2015). The evidence in favor of this hypothesis is the *Is*(*T*) curves for the samples of the upper unit of the Khaipakh formation in which the Curie point is clearly observed close to 680°C (Fig. 4). The pattern of the hysteresis loops (with the absence of saturation in the fields exceeding 1 T) and the curves of normal magnetization (with the absence of saturation in the fields on the order of 1.5 T) support this conclusion (Figs. 5a and 5b). The CLG analysis of the normal magnetization curve (Kruiver et al., 2001; Maxbauer et al., 2016) testifies to the presence of three highly coercive minerals with the median saturation fields $H_{1/2}$ = 154, 440, and 448 mT in the limestone sample of the Upper Khaipakh Formation (Fig. 5c). The array of the rock magnetic parameters supports the conclusion that the main magnetic mineral is hematite with various grain sizes.

The vectors of the medium-temperature component are located on the stereogram in the vicinity of the direction of the present geomagnetic field in the study region (Fig. 11c). These vectors are probably associated with the viscous magnetization that was created in the rocks under the action of the recent geomagnetic field.

The directions of the high-temperature component have a bipolar distribution (Fig.11a). On the stereograms, they form a cluster of the directions with the northeasterly declination and moderate inclination and a cluster of the directions with the southwesterly declination and gentle negative and positive inclinations.

Several vectors are located on the arc of a great circle passing through the above clusters and the direction of the present geomagnetic field in the study area (Fig. 11a). This allows us to conclude that the high-temperature magnetization component in some samples is to a varying extent contaminated by the recent viscous component. This type of the samples was not considered in the calculations of the mean direction of the high-temperature magnetization component (Fig. 11b).

The mean direction of magnetization was calculated both by averaging the high-temperature directions in all samples and by averaging the directions isolated separately in each of the five sites (Fig. 11d).

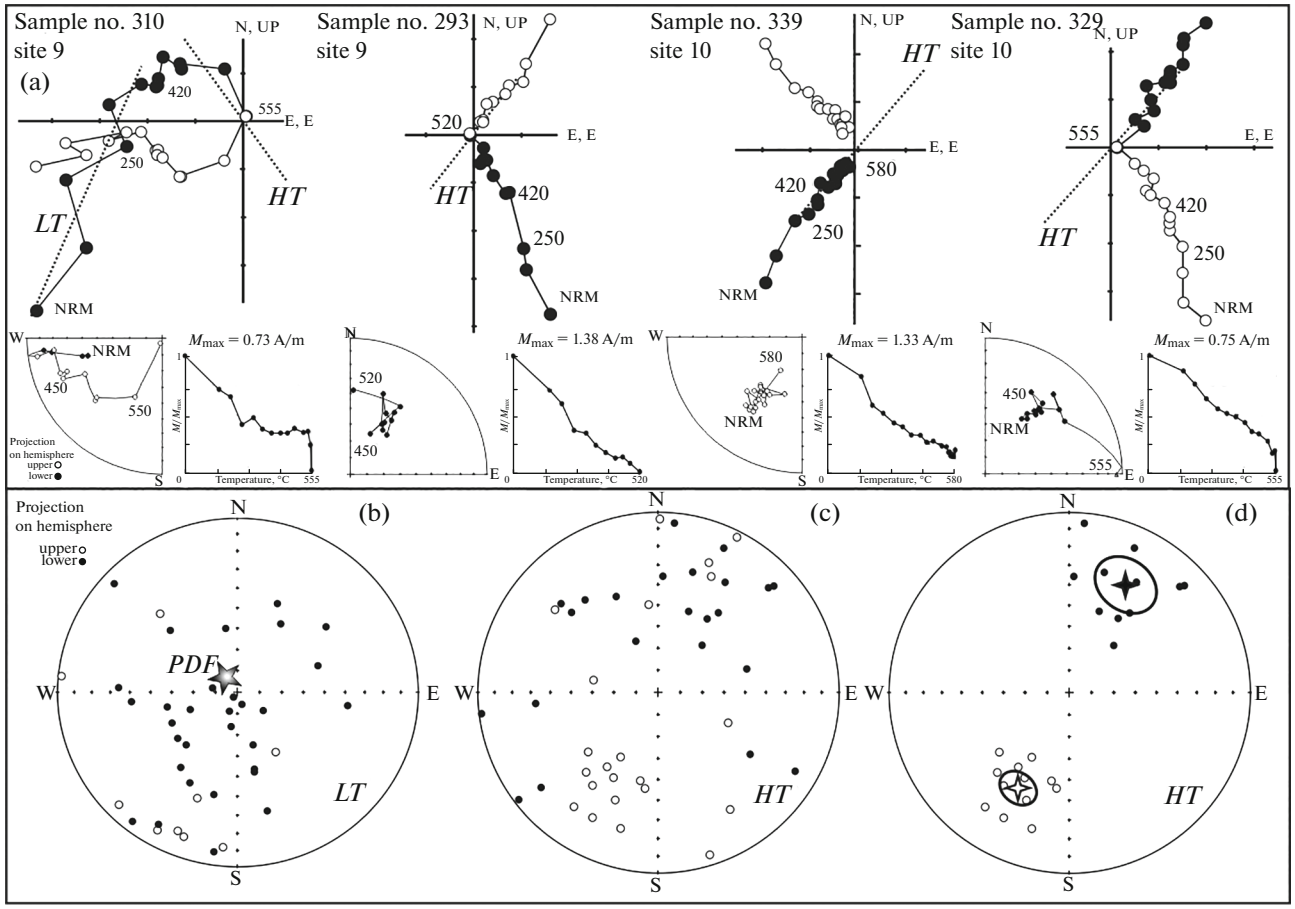


Fig. 9. Demagnetization results of Udzha Uplift dolerite sill samples (sites 9 and 10): (a) representative Zijderveld diagrams, stereograms, and demagnetization diagrams; (b) directions of low-temperature magnetization components (*LT*); (c) directions of high-temperature magnetization components (*HT*); (d) directions of high-temperature magnetization components used in calculation of mean direction. *PDF* is direction of present magnetic field in study region. Diagrams are presented for stratigraphic coordinate system.

OPTICAL, ELECTRON-MICROSCOPIC, AND MICROPROBE STUDIES

For identifying the indications of primary or secondary origin of the magnetic minerals with respect to the formation time of the considered rocks, we carried out optical, electron microscopic and microprobe studies.

Igneous and Volcanic-Sedimentary Rocks

Based on the results of the conducted studies, the structures of heterophase oxidation (HPO) were identified in titanomagnetite grains in tuffs of the Unguokhtakh Formation and in dolerites of sites 9 and 10. The HPO structures manifest itself in the exsolution of primary titanomagnetite grains into lamellae of ilmenite and magnetite (Fig. 12, samples 212 and 284). These structures are indicators of the fact that magnetite has been formed at high temperatures during melt crystallization and, therefore, it may carry primary magnetization reflecting the direction of the geomagnetic field during the formation of rocks (Dunlop and Özdemir, 1997). The preservation of the HPO structures also suggests the absence of significant transformations of the rock throughout the period of its existence.

The Unguokhtakh tuffs contain structures that may indicate tuff formation at very high temperatures. The example of this structure is presented in Fig. 12. Here, the photo of sample 212 shows a lithoclast with diffuse boundaries testifying to its secondary melting. This type of the structures is characteristic of tuffs that form from the incandescent products (including lithoclasts) of an explosive eruption.

The study of thin sections of basalts from site 6 revealed skeletal crystals of titanomagnetite (Fig. 12, sample 281) which are formed at rapid crystallization of the melt at the moment of formation of the rock. Extremely small (<1 μm) titanomagnetite grains with HPO structures (Fig. 12, sample 281) also occur. These facts suggest that we are dealing with primary

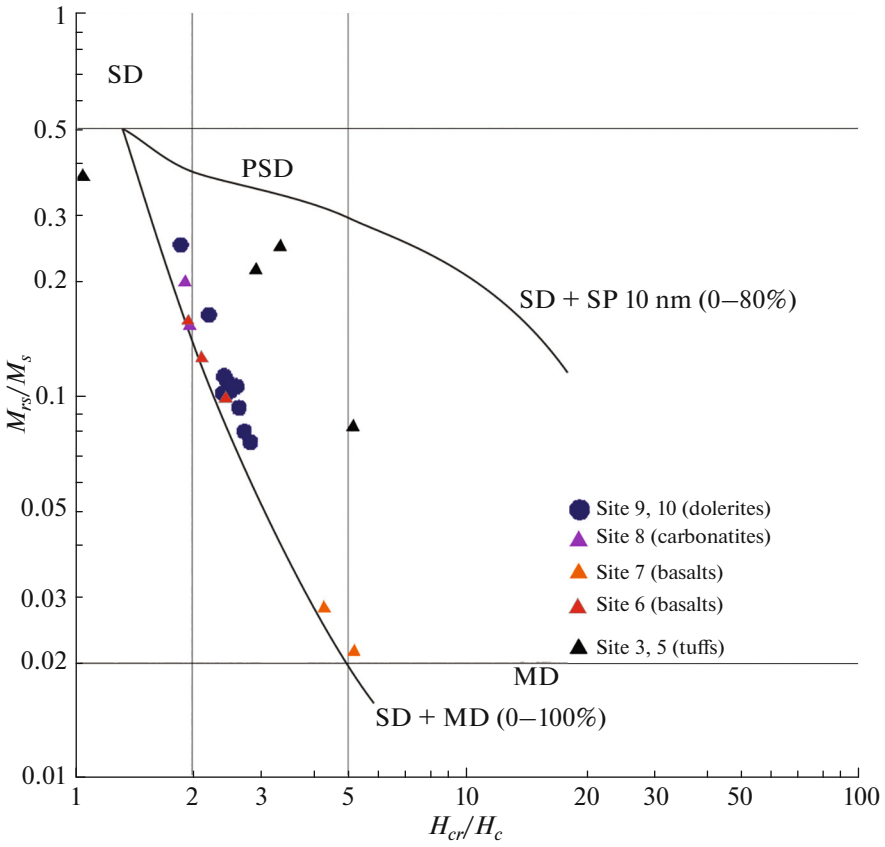


Fig. 10. Day–Dunlop diagram (Day et al., 1977; Dunlop, 2002a; 2002b). SD, PSD, MD are regions of single-domain, pseudo-single-domain, and multi-domain grains, respectively.

titanomagnetite grains that were formed at the moment of rock crystallization.

Basalt samples from site 7 have numerous macroscopically observed secondary transformations (strong ferruginization of samples) and, therefore, were not subjected to the detailed microprobe study. In the carbonatite dike (site 8), submicron formations with a Ti-depleted titanomagnete composition of unknown origin are detected.

Sedimentary Rocks

The microprobe and optical studies have shown that the sandstones of the Udzha Formation contain two generations of hematite (Fig. 12, sample 149a). The first generation is represented by detrital rounded grains and oolites which are most likely to have been formed directly during the sedimentogenesis. The second generation of hematite (pigment hematite) is represented by small, up to 10 μm flakes which border large grains of quartz and first-generation hematite and constitute sandstone cement.

In quartz grains, the incorporation (indentation) structures and the indications of their recrystallization were detected (Fig. 12, sample 149). These signs indicate that the sandstones of the Udzha Formation underwent transformations of the late diagenesis stage of. We suppose that the redistribution of iron from detrital hematite grains into interstitions with the subsequent formation of pigment hematite took place exactly at this stage. Apparently, radial and concentric zones of a "turbid" photo tone on the "bright" grains of detrital hematite (Fig. 12, sample 149b) are the traces of this process.

The electron microscopic study of the siltstones of the Unguokhtakh Formation revealed titanomagnetite grains probably having detrital origin with a size of up to 10 μm (Fig. 12, sample 254).

CONSTRAINTS ON THE FORMATION TIME OF ANCIENT MAGNETIZATION COMPONENT ISOLATED IN THE STUDIED ROCKS

The results of the paleomagnetic analysis show that all the studied objects contain high-temperature magnetization components (*HT*); besides, in the rocks of the Unguokhtakh and Udzha formations, also the medium-temperature components *MTW* and *MTE* (Udzha formation) are isolated.

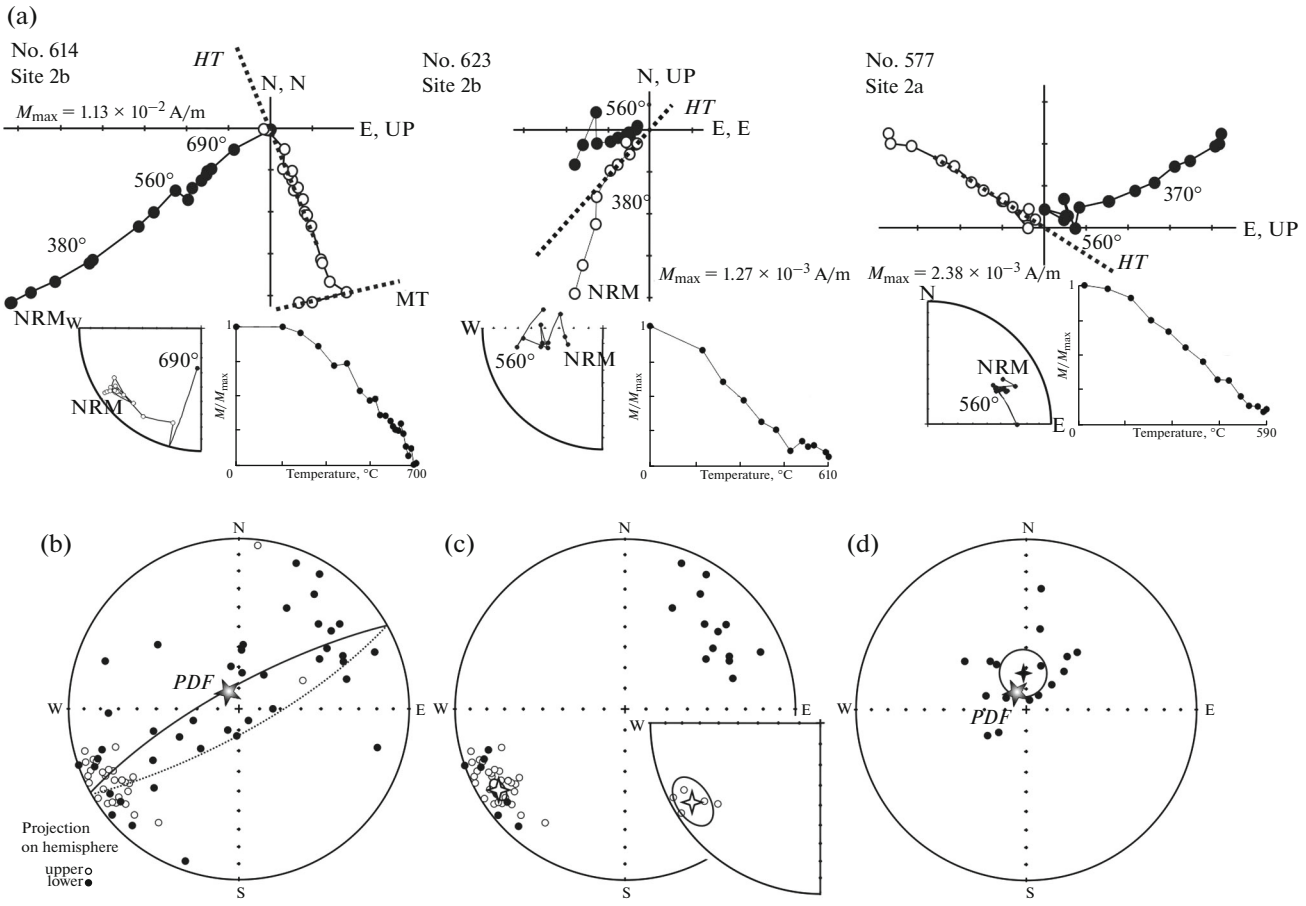


Fig. 11. Results of paleomagnetic study of limestones of Olenek Uplift Khaipakh Formation: (a) representative Zijderveld diagrams, stereograms, and demagnetization diagrams; (b) directions of high-temperature magnetization components at all five sites; (c) magnetization directions not contaminated by present field included in statistics; (d) directions of low-temperature magnetization components; (e) site mean magnetization directions. *PDF* is direction of present magnetic field in study region. Diagrams are presented in geographic coordinate system.

The paleomagnetic pole calculated from the direction of the *MTW* component (Table 2) is located in the region of the Early to Mid-Cambrian paleomagnetic directions of the Siberian Platform (Khramov et al., 1982; Pavlov et al., 2018) (Fig. 13). However, as of now, no manifestations of the geological processes that could serve as a source of remagnetization of the studied rocks in the Early–Middle Cambrian have been described in the territory of the Udzha Uplift. At the same time, the signs of Early Cambrian magmatic activation were noted in the northeast of the Siberian Platform. For example, Bowring et al. (1993) described the Early Cambrian intrusive bodies and diatremes in the Olenek Uplift (*Ob"yasnitel'naya*..., 1983); the occurrences of the bimodal Early Cambrian magmatism in the lower reaches of Lena River were reported in (Bowring et al., 1993; Prokopiev et al., 2016). The probable high rate of the Early Cambrian endogenous activity in this region is suggested by the extensive occurrence of detrital zircons with an age of 540–520 Ma in the north and northeast of the Siberian Platform (Vishnevskaya et al., 2017; unpublished data of the authors) whose source is most likely to be the above diatremes.

The mean direction of the *MTE* component lies close to the great circle passing through the mean directions of the normal and reversed polarity vectors of the *HT* component of the Udzha Formation (Fig. 3). This fact may indicate the contamination of the primary *HT* component by the *MTE* component suggesting the secondary origin of the latter. The pole calculated from the *MTE* component is located in the vicinity of the Early Carboniferous segment of the APWP curve for the Siberian Platform (Torsvik et al., 2012) (Fig. 13). The close ages (~360–330 Ma) in the northern Siberian Platform are noted among the K–Ar and U–Pb isotope datings of the intrusive bodies on the eastern slope of the Anabar Shield and within the Udzha and Olenek uplifts. According to the data of (*Ob"yasnitel'naya*..., 1983; 2013), in this region, magmatic bodies of the Molodo–Udzha complex with isotopic ages of ~420–300 Ma are present. Besides, on

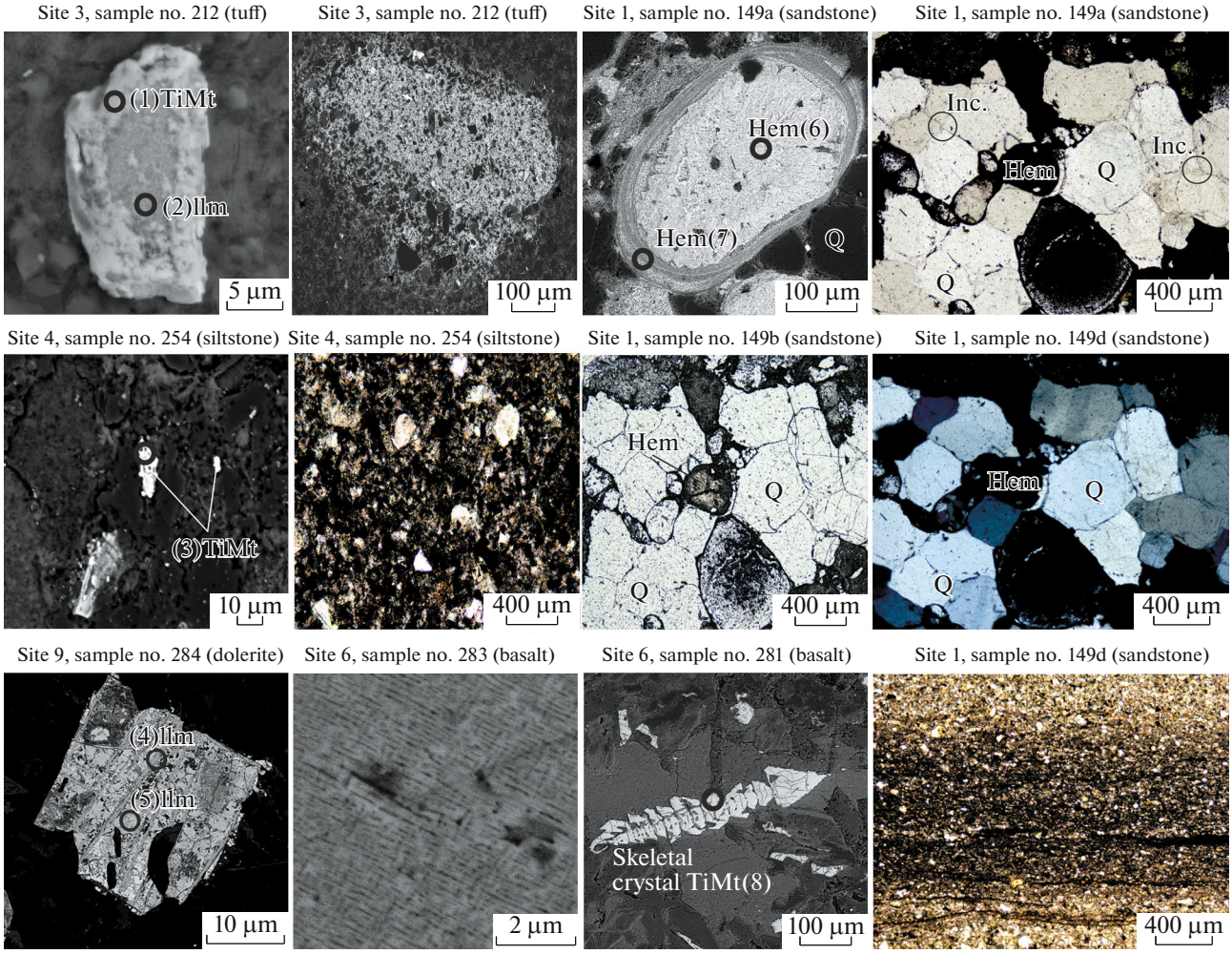


Fig. 12. Results of microprobe (Table 3) and optical study of thin sections of sedimentary and igneous rocks of Udzha uplift. No. 212 is clastic titanomagnetite of Unguokhtakh Formation with exsolution structures. No. 212a is submelted lithoclast in tuff of Unguokhtakh Formation. No. 281 in skeletal titanomagnetite crystal in basalts of site 6. No. 284 is HFO structures in titanomagnetite crystal in dolerites of site 9. No. 283 is fine HFO lattice in titanomagnetite crystal of basalts of site 6. No. 149a is hematite oolite with quartz grains and hematite cement in Udzha Formation sandstone. No. 149b is general view of thin section of Udzha Formation sandstone, reflected light image. No. 149c is transmitted-light view of No. 149b, parallel nicols. No. 149d is transmitted-light view of No. 149b, crossed nicols. No. 254 is clastic titanomagnetite in Unguokhtakh Formation siltstones. No. 254 is general view of Unguokhtakh Formation siltstone in transmitted light with parallel nicols. Inc. means incorporation structures of quartz grains. Numbers in parentheses indicate analysis number in Table 3. Abbreviations Ilm, TiMt, Hem, and Q stand for ilmenite, titanomagnetite, hematite, and quartz, respectively.

the western slope of the Olenek uplift, the diatremes of the Toluop–Merchimden kimberlite complex with the Rb-Sr isotopic ages of 341–374 Ma are documented (*Ob"yasnitel'naya*..., 1983). These data definitely indicate that the geological processes that existed in this region in the Early Carboniferous could lead to the formation of a secondary magnetization component in sedimentary rocks of the Udzha uplift. Another probable source of remagnetization could have been the formation of the complex of alkaline and alkaline-basic intrusions which are mainly represented in this region by the Tomtor massif. However, the isotope dating of this massif known to date (U–Pb age 701–675 Ma and U–Pb age 414–387 Ma) (Vladykin et al., 2014) do not support this hypothesis.

The high-temperature *HT* components of the Unguokhtakh and Udzha formations are bipolar, the reversal test is positive for the Unguokhtakh (γ/γ_c = 0.8°/10.3°, class C) and negative (γ/γ_c = 34.7°/17.7°) (McFadden and McElhinny, 1990) for Udzha Formation. The directions of the high-temperature components of these formations are noticeably different whereas a common secondary *MTW* component is identified in both formations. This testifies to the primary origin of the detected high-temperature components and to the absence of the signs of regional remagnetization.

Table 2. Directions of magnetization components isolated in rocks of Udzha and Olenek uplifts

		Geographic coordinate system				Stratigraphic coordinate system				Paleopole		
Site	*n*	*D*°	*I*°	*K*	α95°	*D*°	*I*°	*K*	α95°	Φ°	Λ°	*dp*/*dm*, A95°
Khaipakh Formation (Olenek uplift) (c.m.p. coordinates N 71.1°, E 124.1°)												
HT component												
2*a*	15	231.7	−24.5	14.0	10.6	–	–	–	–			
2*b*	6	243.9	−11.7	64.6	8.4	–	–	–	–			
2*c*	9	243.1	−3.7	32.3	9.2	–	–	–	–			
S14OLD	10	236.0	−19.1	68.0	5.9	–	–	–	–			
S15OLD	10	237.1	−2.9	29.1	9.1	–	–	–	–			
Normal polarity	37	239.2	−8.7	34.0	4.2	–	–	–	–			
Reversed polarity	13	48.3	28.1	18.7	9.8	–	–	–	–			
Sample mean	**50**	**237.4**	**−13.8**	**19.5**	**4.7**	**–**	**–**	**–**	**–**	**−16.8**	**63.4**	**2.5/4.8, 3.4**
Site mean	5	238.5	−12.4	57.8	10.1	–	–	–	–			
Udzha Formation (Udzha uplift) (c.m.p. coordinates N 70.8°, E 117°)												
MTW component												
1, 2	20	344.0	46.4	10.5	10.6	343.9	46.0	10.1	10.8			
MTE component												
1, 2	16	131.2	47.9	8.5	13.5	130.0	45.4	9.2	12.8	13.8	161.7	10.3/16.3, 13.0
HT component												
1	11	233.7	14.5	19.6	10.6	234.8	8.6	19.8	10.5			
2	27	226.6	−1.6	9.1	9.7	226.4	3.8	9.1	9.7			
Normal polarity	13	52.8	22.5	20.9	9.3	55.5	16.6	22.8	8.9			
Reversed polarity	25	226.6	16.3	19.6	6.7	225.5	16.7	16.7	7.3			
Sample mean	**38**	**228.7**	**3.2**	**10.0**	**7.7**	**228.9**	**5.3**	**10.7**	**7.4**	**−9.9**	**67.2**	**3.7/7.4, 5.3**
Exposure mean	2	229.6	−3.1	–	–	230.5	0.1	–	–			
Unguokhtakh formation (Udzha uplift) (c.m.p. coordinates N 70.9°, E 116.9°)												
MTW component												
3, 4, 5	18	330.7	39.6	11.3	10.8	337.9	45.2	11.1	10.9			
HT component												
3 (tuff)	8	214.5	−26.9	27.3	10.8	220.9	−38.4	27.3	10.8			
4 (siltstone)	23	37.4	27.9	23.6	6.4	41.0	22.0	22.1	6.6			
5 (tuff)	6	220.3	−22.6	21.2	14.9	–	–	–	–			
Normal polarity	24	37.3	27.9	24.6	6.1	41.0	22.7	22.2	6.4			
Reversed polarity	12	217.3	−27.2	27.2	8.5	220.9	−32.9	20.4	9.8			
Sample mean	**36**	**217.3**	**−27.6**	**26.1**	**4.8**	**220.9**	**−26.1**	**20.7**	**5.4**	**−27.7**	**71.0**	**3.2/5.8, 4.3**
Exposure mean	3	217.4	−25.8	443.7	5.9	220.7	−27.7	76.0	14.2			
MTW component (*Udzha and Unguokhtakh formations*, *c.m.p. coordinates N 70.8°, E 117.0°*)												
1, 2, 3, 4, 5	38	–	–	–	–	341	45.6	10.8	7.4	−44.9	141.2	6.0/9.4, 7.5
Magmatic bodies of Udzha uplift (c.m.p. coordinates N 71.0°, E 117.0°)												
HT component												
6 (basalt)	7	197.4	−33.5	42.2	9.4	–	–	–	–	−36.3	96.4	6.1/10.7, 8.1
7 (basalt)	7	220.9	−37.4	33.4	10.6	–	–	–	–	−37.9	69.7	6.6/11.9, 8.9
8 (carbonatite)	7	29.1	10.8	79.7	6.8	–	–	–	–	−21.9	85.5	3.5/6.9, 4.9
9 (dolerite)	6R+3N	200.5	−28.1	18.0	12.5	204.2	−42.2	18.0	12.5			
10 (dolerite)	6R+4N	208.5	−17.7	20.9	10.8	209.9	−37.0	20.2	11.0			

Table 2. (Contd.)

		Geographic coordinate system				Stratigraphic coordinate system				Paleopole		
Site	*n*	*D*°	*I*°	*K*	α95°	*D*°	*I*°	*K*	α95°	Φ°	Λ°	*dp*/*dm*, A95°
Normal polarity	7	24.1	20.8	10.9	19.2	25.7	38.6	10.8	19.2			
Reversed polarity	12	205.3	–23.7	25.5	8.8	208.2	–39.9	30.3	8.1			
Sample mean	19	207.6	–25.4	19.1	7.9	211.0	–41.9	21.3	7.4	–39.9	79.2	5.9/9.1, 7.1
Mean pole	**4**	–	–	–	–	–	–	–	–	**–34.4**	**82.9**	**14.0**

MTW, *MTE* are medium-temperature magnetization components (explanations are in text); *HT* is high-temperature magnetization component; c.m.p.—conditional midpoint; *n* is number of vectors used in statistics; *D* is declination; *I* is inclination; *K* is concentration parameter; α95 is radius of 95% confidence ellipse; Φ is latitude, Λ is longitude; A95 is radius of 95% confidence ellipse for paleopole. Directions of components for objects with horizontal bedding are presented in geographic coordinate system.

Table 3. Chemical composition of magnetic minerals in rocks of Udzha uplift based on results of microprobe analysis

No.	Sample no,.	Site		O	Al	Si	Ca	Ti	V	Mn	Fe	Zn	Total
1	212	3	TiMt	32.61	–	–	–	3.83	–	–	63.56	–	100.00
2	212	3	Ilm	38.16	–	–	–	32.86	–	–	25.59	–	96.61
3	254	4	TiMt	27.97	–	0.34	–	4.39	–	–	66.13	–	98.83
4	284	9	Ilm	29.53	0.99	–	–	12.53	0.61	–	55.64	0.49	99.79
5	284	9	Mt	32.27	–	0.29	0.31	30.42	–	4.77	33.94	–	102.00
6	149	2	Hem	26.89	0.89	1.49	–	–	–	–	70.47	–	99.74
7	149	2	Hem	30.95	3.28	4.76	–	–	–	–	59.97	–	98.96
8	281	6	TiMt	28.06	0.62	–	0.24	9.29	0.55	0.33	57.74	–	96.83

Abbreviations Ilm, TiMt, and Hem stand for ilmenite, titanomagnetite, and hematite, respectively.

The magnetomineralogical and microscopic studies show that the rocks of the Unguokhtakh Formation contain hematite and detrital grains of magnetite and titanomagnetite with high-temperature exsolution structures. The evidence has been found that tuffs of the Unguokhtakh Formation remained heated to very high temperatures during their deposition and, hence, they can possess thermoremanent magnetization. The direction of the *HT* component in the rocks of the Unguokhtakh Formation does not depend on the magnetic carrier mineral, which is yet another argument in favor of the primary origin of the magnetization. Based on the above arguments, we believe that the high-temperature magnetization components identified in the rocks of the Unguokhtakh Formation are primary and the time of their formation corresponds to the time of the formation of the rocks.

The magnetomineralogical and microscopic studies of sandstones of the Udzha Formation indicate the presence of primary detrital and secondary pigment hematite. The bipolar distribution of the direction of the *HT* component in pebbles which coincides with the direction of this component in sandstones suggests that the magnetization was formed after lithification of conglomerates. The negative pebble test can mean that the magnetization in the Udzha sandstones could have been formed during the diagenesis and can be associated with authigenic hematite. The bipolar distribution of the directions of this component (Fig. 3), its stability relative to the *MTE* component, and the fact that the pole calculated from the direction of this magnetization does not coincide with the younger known poles (Fig. 13) suggest that the formation time of the high-temperature magnetization component is close to the formation time of the Udzha Formation sandstones.

The negative reversal test for the rocks of the Udzha Formation indicates that the revealed directions of the characteristic component are substantially contaminated by the superimposed secondary components. However, a simple modeling conducted in our study shows that in the case of averaging of the bipolar magnetization components, the influence of the secondary component typically decreases and even vanishes in some special cases. Thus, the obtained mean direction will insignificantly deviate from the true dikrection.

The high-temperature characteristic component in the rocks of the Upper Khaipakh Formation is also represented by the vectors of normal and reversed polarity. While the declinations of the respective mean directions differ by almost 180°, their inclinations are not symmetrical about zero (–8.7° against 28.1°). The reversal test conducted for these directions is negative

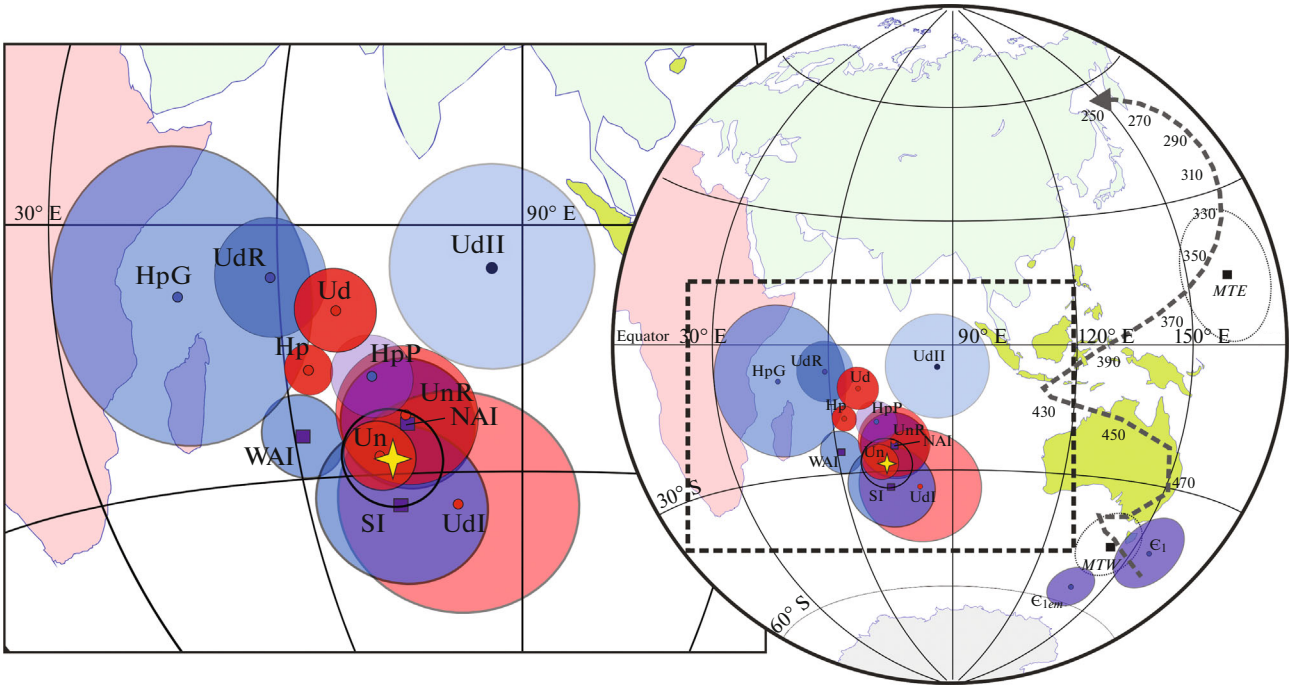


Fig. 13. Poles obtained in this work (red), their comparison with poles obtained by authors of previous studies (blue). Pole codes are indicated in Table 4. Boxes show poles presented in (Evans et al., 2016) and poles calculated from *MTW* and *MTE* components. Yellow asterisk is averaged paleomagnetic pole of age ~1500 Ma proposed in this work. Dotted line shows APWP curve for Paleozoic of Siberian Platform (Torsvik et al., 2012).

($\gamma/\gamma_c = 22.0°/8.9°$) (McFadden and McElhinny, 1990). This is understandable if we assume that the recent component has not been not completely removed by thermal demagnetization. As in the case of the formations considered above, the paleomagnetic pole calculated for the *HT* component of the Upper Khaipakh Formation noticeably differs from all younger poles known for the Siberian Platform.

Unfortunately, the geological situation precludes from conducting any additional tests whose results could constrain the formation time of the *HT* component of the upper unit of the Khaipakh formation. Nevertheless, the fact that the hematite grains of different sizes, which can also have different origin, carry a similar paleomagnetic signal may testify in favor of the primary origin of the magnetization.

The coincidence of the magnetization directions for hematite grains of different sizes/generations, the presence of the vectors of normal and reversed polarity, and the difference of the direction of the high-temperature component from the younger Siberian paleomagnetic directions suggest that the age of the high-temperature magnetization component of the upper unit of the Khaipakh formation is close to the time of formation of the rock.

The directions of the *HT* magnetization components of the studied magmatic bodies are close among each other and are also close to the mean direction of the *HT* component of the Unguokhtakh Formation. This can mean a close age of the formation of these rocks.

The high-temperature magnetization component in the dolerite sill (sites 9 and 10) has a bipolar distribution ($\gamma/\gamma_c = 2.3°/16.8°$, class C) (McFadden and McElhinny, 1990) which can barely be explained otherwise as resulting from magnetization self-reversal. This self-reversal can occur either during the cooling of the rock or during the subsequent transformations of the initial magmatic minerals. In both cases, the pole calculated from the isolated directions should reflect the direction of the geomagnetic field at the time of the emplacement of magmatic bodies.

In the tuffs of the Unguokhtakh Formation, we detected single zircon grains presumably having primary magmatic origin. The U–Pb age of these zircons is 1436 ± 72, 1480 ± 58, and 1527 ± 75 Ma (authors' unpublished data). Because of the extremely scarce number of grains, these determinations cannot be directly correlated to the age of the rock; however, they can be interpreted as an indication that the rocks under study could have been formed close to ~1500 Ma. If this assumption is the case, then the coincidence of the poles of the studied magmatic bodies and the rocks of the Unguokhtakh Formation with the poles obtained from the intrusions of the Anabar and Olenek uplifts having an age of ~1470–1500 Ma is an argument in favor of the primary origin of the magnetization identified in the studied bodies (Fig. 13 , Table 4).

Table 4. Paleomagnetic poles of Siberian platform obtained in this work and poles obtained by authors of previous studies

Object	Code	Age, Ma	Φ°	Λ°	A95(°)	Source
West Anabar Intrusions	WAI	1503 ± 2	–25	61	5	(Evans et al., 2016)
North Anabar Intrusions	NAI	1483 ± 17	–24	75	8	(Evans et al., 2016)
Sill of Sololi River	Sl	1473 ± 24	–34	73	10	(Wingate et al., 2009)
Unguokhtakh Formation	UnR*		–23	75	9	(Rodionov, 1984)
Unguokhtakh Formation	Un		–28	71	4	This work
Stage I Udzha magmatism	UdI		–34	83	14	This work
Mesoproterozoic intrusions of northern Siberia ($N = 5$; WAI + NAI + Sl + Un + UdI)	NSI	≈1500	–28	73	6	This work
Udzha Formation	UdR	R3	–6	59	7	(Rodionov, 1984)
Udzha Formation	Ud	1500–1386	–10	67	5	This work
Khaipakh Formation	Hp	<1030	–17	63	3	This work
Khaipakh Formation	HpP	900–800	–18	71	5	(Pavlov et al., 2015)
Khaipakh Formation	HpG	<1030	–8	47	17	(Gurevich, 1983)
Stage II Udzha magmatism	UdII	≈1385	–5	85	11	(Malyshev et al., 2018)
–	$Є_1$	$Є_1$	–44	157	8	(Khramov et al., 1982)
Emyaksin Formation	$Є_{1em}$	$Є_1$	–56	138	5	(Pavlov et al., 2018)

* Pole UnR is not used in calculation of the mean direction of Mesoproterozoic intrusions in northern Siberia. Paleopole parameters: Φ is latitude, Λ is longitude; A95 is radius of 95% confidence ellipse.

DISCUSSION

Based on the conducted studies, we identified the characteristic magnetization components in the rocks of the upper unit of the Khaipakh formation of the Olenek Uplift, the Unguokhtakh and Udzha formations of the Udzha Uplift and magmatic bodies of the first stage of magmatism of the Udzha Uplift (Table 2).

The obtained paleomagnetic and rock magnetic data suggest that these magnetization components were created during or shortly after the formation of the corresponding rocks and that they reflect the direction of the geomagnetic field at the time of their formation. For each the studied igneous body, paleomagnetic poles were calculated from the mean directions of the characteristic components (Table 2). The calculated poles are close among each other and close to the pole corresponding to the characteristic component of the Unguokhtakh Formation. We also note that all these poles are located in the vicinity of the pole of the Unguokhtakh Formation proposed by our predecessors (Rodionov, 1984) based on a more limited statistical material and using less intense demagnetization (Fig. 13, Table 4).

Moreover, these poles lie in the same region as the paleomagnetic poles obtained from the dated intrusive bodies of the Anabar and Olenek uplifts which pertain to the Kuonamka Large Igneous Province (1513 ± 51, 1503 ± 2, 1473 ± 24 and 1483 ± 17 Ma) (Veselovskiy et al., 2006; Evans et al., 2016). This fact suggests that the above objects were formed close in time and, probably, within the time span of the magmatic event that led to the formation of the Kuonamka Igneous Province about 1500 Ma ago.

It appears that with the current knowledge about this magmatic episode, there is no point in trying to separate the paleomagnetic poles known for this episode from the poles obtained from the contemporaneous formations of the Udzha uplift in terms of their age. Therefore, as the pole for Siberia with an age of ~1500 Ma, we propose to use the paleomagnetic pole calculated as a mean over all the poles obtained from the studied igneous rocks, sedimentary rocks of the Unguokhtakh Formation, and the objects of the Kuonamka Igneous Province (Table 4).

The paleomagnetic pole calculated from the characteristic component isolated in the rocks of the Udzha Formation of the Udzha uplift coincides, within the confidence ellipse, with the previously obtained pole (Rodionov, 1984) (Table 4). However, we propose to consider the new pole a more reliable as it is based on a larger statistics and obtained with complete demagnetization of NRM.

The new paleomagnetic pole of the Udzha Formation is close to that obtained for the upper unit of the Khaipakh formation of the Olenek Uplift; however, the confidence ellipses of these poles do not overlap and the difference in their coordinates is statistically significant (γ/γ_c = 7.4°/4.8° (Debiche and Watson, 1995)) (Table 4, Fig. 13). The misalignment of the poles contradicts the correlation of the Udzha Forma-

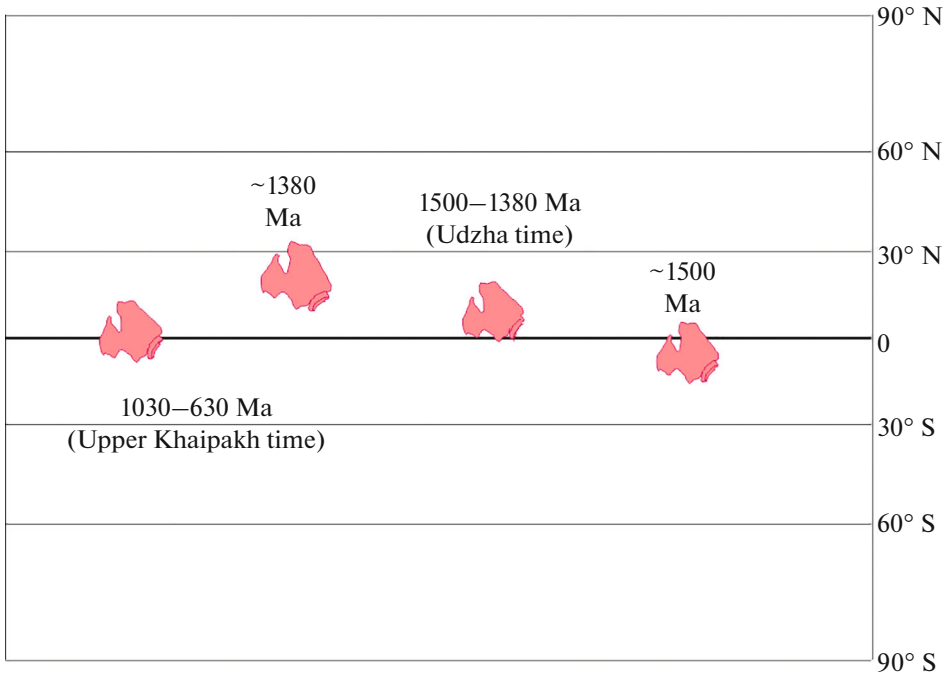


Fig. 14. Reconstructions of latitudinal position of Siberian platform in Mesoproterozoic based on new paleomagnetic data presented in this work.

tion and the upper unit of the Khaipakh formation, which relies on the results of interregional correlation of stromatolite forms (Semikhatov and Serebryakov, 1983).

Thus, our data testify in favor of non-simultaneous formation of the studied sedimentary rocks.

According to the most recent results, the age of the Udzha Formation cannot be younger than 1386 ± 30 Ma (Malyshev et al., 2018). On the other hand, this formation lies stratigraphically higher than the Unguokhtakh Formation whose age, according to the paleomagnetic and isotopic data (see above), is probably about 1500 Ma. Thus, we may conclude that the rocks of the Udzha Formation of the Udzha uplift were formed in the Mesoproterozoic in the interval 1500–1385 Ma.

The recent isotope age determinations for the lower unit of the Khaipakh formation (Rb-Sr, glauconite, 1172 ± 12 and 1112 ± 24 Ma) (Zaitseva et al., 2017) constrain the probable age of the upper unit of the Khaipakh formation from below. The likelihood of resetting of the Rb-Sr system in glauconites during diagenetic changes in rocks has been repeatedly discussed in the literature (Grant et al., 1984; McDougall, 1977); and recently, the "rejuvenation" of the age of globular layered silicates in the rocks of the Arymas and Debengda formations due to the postdiagenetic transformations was demonstrated in (Zaitseva et al., 2018). However, as of now there is no evidence in favor of a probable "rejuvenation" of the Rb–Sr age of the lower unit of the Khaipakh formation. Thus, the current data do not contradict the opinion that the upper unit of the Khaipakh formation has the Neoproterozoic age (Semikhatov and Serebryakov, 1983; Semikhatov et al., 2000; Zaitseva et al., 2017).

Based on the obtained data, we can determine the paleogeographic position of the Siberian craton and the pattern of its movements in the time interval ~1500–1380 Ma ago. For doing this, we should however first of all opt for the polarity of the paleomagnetic directions that will be used for this purpose. Following (Pavlov et al., 2015; 2019), based on the displacement minimization principle, and taking into account the position of the Totta paleomagnetic pole (Pavlov, 1994) which is closest in age among the known Late Proterozoic poles, we choose the so-called "Indian Ocean" polarity option according to which the Late Precambrian northern paleomagnetic poles for Siberia are located in the eastern hemisphere in the Indian Ocean.

With this polarity option, our data mean that ~1500 Ma ago, the Siberian Craton was almost entirely located at the equatorial latitudes of the southern hemisphere (Fig. 14). Throughout the considered time period, the Siberian craton was slowly moving (by ~2° over 10 Ma) to the north. By the Udzha time, the craton had almost completely crossed the equator, and by the time of ~1380 Ma ago, the southernmost part of the crater had reached the tropical region and was located in the vicinity of 30° N. All the time through, the craton was oriented by its southern (in present coordinates) part to the north and did not experience significant rotations relative to the merid-

ian. In the Late Khaipakh time, the Siberian platform has got again located in the equatorial region.

CONCLUSIONS

Based on the study of the sedimentary and igneous rocks of the Udzha and Olenek uplifts, we obtained the following results.

—Besides the intrusive bodies with an age of ~1380 Ma (Malyshev et al., 2018), in the territory of the Udzha uplift are also the magmatic bodies with an age of ~1500 Ma which are most likely to belong to the Kuonamka Large Igneous Province (Ernst et al., 2016).

—It is shown that the paleomagnetic poles of the upper unit of the Khaipakh formation of the Olenek uplift and the poles of Udzha Formation of the Udzha uplift misalign which suggests that they were formed at different times. This contradicts the accepted correlation scheme of these formations (Semikhatov and Serebryakov, 1983).

—The new paleomagnetic poles are obtained for the formation times of the Upper Khaipakh Subormation and Udzha and Unguokhtakh formations of the Siberian Platform.

—The paleomagnetic pole that averages the paleomagnetic determinations obtained from the sedimentary and igneous rocks of the Udzha uplift and from the magmatic bodies of the Kuonamka magmatic province of the Siberian platform (~1500 Ma) is proposed.

—Based on new paleomagnetic data it is shown that in the time interval from ~1500 to 1385 Ma and in the Late Khaipakh time, the Siberian platform was located in the equatorial latitudes initially of the southern and subsequently of the northern hemisphere, and was oriented to the north by its present southwestern margin (Fig. 14).

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